# The complete supported-interval rank step

Human Compression project: bounded L044 proof component

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le A_n$, $n \ge 12$, be transitive and a string C-group, with all maximal omissions intransitive. Suppose the unique eligible interior $i$-edge joins its two omission orbits $A, B$, and the active label sets on these two orbits are each single intervals. As proved in the accepted supported-interval lemma, after interchanging the orbits the actual groups $L = G_{<i}$ and $R = G_{>i}$ are supported on and transitive on $A$ and $B$, respectively. Their even string C-markings have connected diagrams, intransitive omissions and endpoint anchors. Assume the alternating rank bound at every strictly smaller degree at least 12.

**Proposition 1.** *Unless the actual first three or last three generators are one of the two degree-six $A_5$ markings of source Table 2, rows (2)–(3), up to full string duality, one has*

$$r \le \left\lfloor \frac{n-1}{2} \right\rfloor .$$

*Proof.* Suppose the asserted rank bound fails. Write $m_A = |A|, m_B = |B|$ and let $d_A, d_B$ be the two interval ranks, so $r = d_A + d_B + 1$. Both degrees are at least four, since a nontrivial even involution cannot be supported on fewer points. For a side of degree $m < 12$, the accepted complete small anchored-imprimitive bound gives $d \le \lfloor (m-2)/2 \rfloor$ in the imprimitive case. In the primitive case, the accepted proper-interval exclusion leaves the same bound unless the marking is source Table 2 row (1), (2) or (3). Rows (2)–(3) are the excluded actual three-generator prefixes or suffixes. Row (1) is the degree-five dihedral head: the complete dihedral attachment proposition, with its explicit six-choice normalization, gives the required global rank bound or exactly those excluded prefixes. Thus this case is impossible under our current contrary assumptions.

For a side of degree $m \ge 12$ which is proper in its natural alternating group, the accepted primitive bound and the complete anchored-imprimitive bound again give $d \le \lfloor (m-2)/2 \rfloor$. For a full natural alternating side the explicit smaller-degree induction hypothesis gives $d \le \lfloor (m-1)/2 \rfloor$. Consequently, unless both sides are full natural alternating groups of degrees at least 12,

$$r \le \frac{m_A - 2}{2} + \frac{m_B - 1}{2} + 1 = \frac{n-1}{2},$$

a contradiction. (Interchange the sides to put the strict bound first.)

In the remaining case use the accepted faithful even one-point extension lemma. Its full proof writes the crossing generator as $(a\ b)uv$, chooses the even exterior factor, and uses actual ambient IP to show that the one-point projection kernel is trivial. Thus one of the two sides, say $A$, extends to a genuine even string C-group of degree $m_A + 1$ and rank $d_A + 1$, with an endpoint anchor and all omissions intransitive. The complete primitive, anchored-imprimitive and smaller-degree alternating bounds apply to this *faithful* marking and give

$$d_A + 1 \le \left\lfloor \frac{m_A}{2} \right\rfloor , \qquad d_B \le \left\lfloor \frac{m_B - 1}{2} \right\rfloor .$$

Since $m_A + 1 < n$, this induction is acyclic. Adding yields the required rank bound and the final contradiction. □

**Exact source and proof scope.** This consolidates Propositions 5.15–5.17 of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for $A_n$*, arXiv:1605.09173v1, physical page 29. The source's interval-deficit and one-point-extension strategy is reused. All small degrees are now covered using the accepted complete Table 2 and anchored bounds and the new complete dihedral adapter. The full one-point kernel proof remains in the paid common component; no projected IP is assumed. The two exceptional $A_5$ prefixes and all later mixed-support and connected-square source branches remain separate. The induction hypothesis is explicit, so this note alone does not prove the complete L044 theorem or admit an original. No new computation or formal theorem is used.